

CASE STUDY

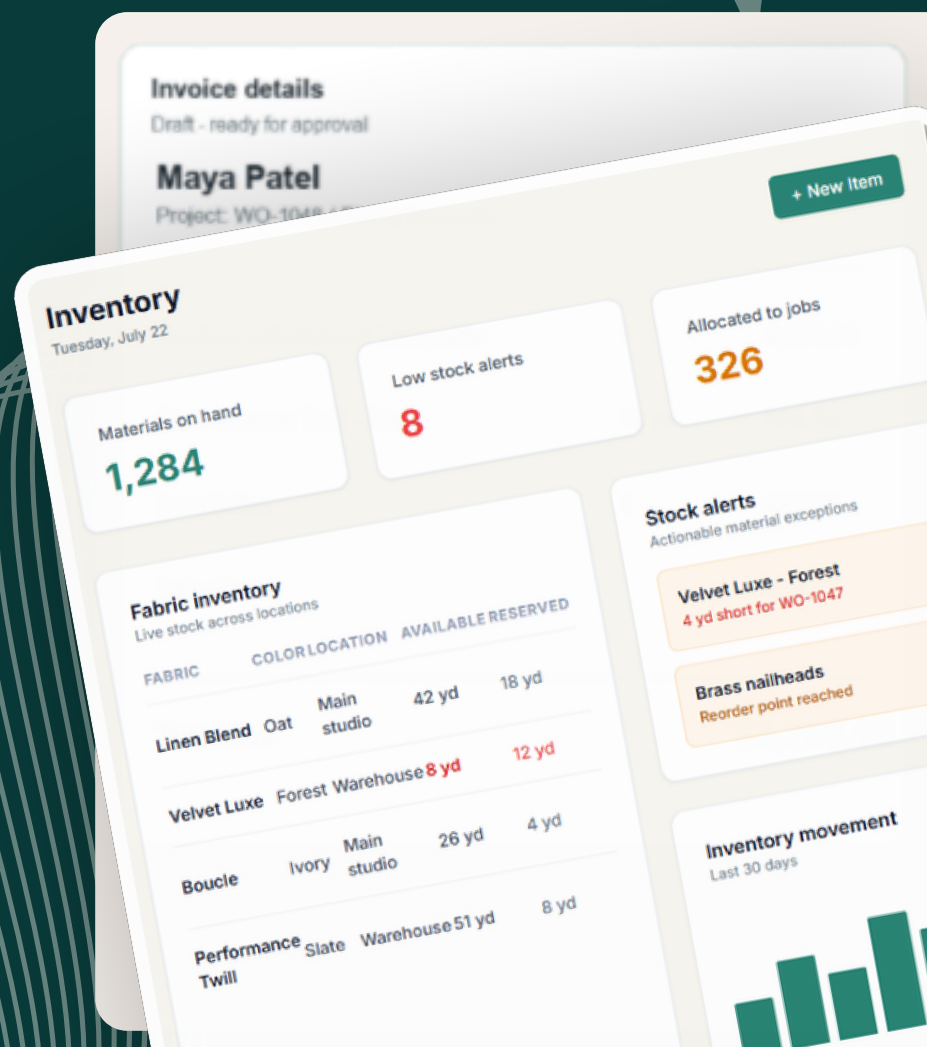


Work Order, Inventory & Invoicing Platform

An integrated operations system built to unify job tracking, material stock, and billing for a custom-manufacturing business

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Problem Statement

The client is an Indian custom-manufacturing business (name withheld for confidentiality) producing made-to-order furniture and upholstered goods for retail and hospitality customers. Production was coordinated through spreadsheets and disconnected trackers, making it difficult to see the live status of jobs, know exactly how much material was on hand versus already committed to a build, or generate an accurate invoice once a job was complete. As order volume grew, this manual approach led to double-booked materials, delayed billing, and limited visibility for managers tracking work across cutting, sewing, upholstery, and quality-check stages.

Client Profile: Indian customer · custom-manufacturing / made-to-order operations · identity kept confidential at the client's request.

Solution Overview

Utkal Labs designed and built a single web application on the MERN stack that connects the entire operational chain — from job creation to material allocation to final billing — so every team works off the same live data instead of separate spreadsheets.

Key Features

- Work order creation & stage tracking
- Automated invoice generation from jobs
- Low-stock & material-shortage alerts
- Live inventory & material reservation
- Guided build configurator with live pricing
- Role-based dashboards for owners & staff

How the Platform Works

1

Work Order Flow

Every job enters the system as a work order carrying the client name, product details, and due date, then moves through defined production stages such as cutting, sewing, upholstery, and quality check. Each order shows a live status — on track, at risk, or ready — along with the owner responsible, so managers can see at a glance which jobs need attention without walking the floor or checking spreadsheets. As a work order progresses, it automatically links forward to the material reservations it consumes and the invoice it will eventually generate.

2

Inventory Flow

The inventory module keeps a live count of raw materials such as fabric and hardware across every storage location, distinguishing stock that is available from stock already reserved against open work orders. When a build is configured or a work order is created, the system automatically checks and reserves the required materials, preventing two jobs from being promised the same limited stock. Low-stock and reorder-point alerts surface automatically as material runs low.

3

Invoicing Flow

Once a work order is marked complete, an invoice can be drafted directly from it in one step, automatically pulling in the labor, materials consumed from inventory, and applicable tax, rather than requiring staff to re-key those figures by hand. Every invoice carries a full audit trail plus payment status and due date, closing the loop from job to material to bill.

Technology Stack

Frontend	React.js, Redux/Context API, HTML5, CSS3 (responsive, component-based UI)
Backend	Node.js, Express.js — REST APIs for work orders, inventory, and invoicing
Database	MongoDB (Mongoose ODM) for jobs, stock ledgers, and invoice records
Auth & Security	JWT-based authentication, role-based access control
Infra & Tooling	Git, REST/JSON APIs, cloud hosting, CI-based deployment

Outcomes

Replacing spreadsheet-based tracking with a connected system gave the client real-time visibility across production, stock, and billing, cutting down the manual reconciliation work that previously slowed every job down.

1

Unified system replacing spreadsheet-based tracking

2

Real-time material visibility, fewer shortage surprises

3

Faster, more accurate invoice turnaround

4

Full traceability from job to material to bill

Product Screens

Work Orders

Overview

Work Orders

Inventory

Build Configurator

Invoicing

Reports

Work Orders

Tuesday, July 22

New +

12 ACTIVE 3 DUE TODAY

Search work orders, clients, or project numbers Filters

WORK ORDER	CLIENT	STAGE	DUE	OWNER	STATUS
WO-1048 Dining banquette - custom	Maya Patel	Cutting	Jul 24	R. James	On track
WO-1047 Dining banquette - custom	Easton Home	Sewing	Jul 23	S. Liu	At risk
WO-1042 Dining banquette - custom	Ava Brooks	Upholstery	Jul 25	M. Jones	On track
WO-1039 Dining banquette - custom	Nolan Hotel	Quality check	Jul 22	R. James	Ready

Inventory

Overview

Work Orders

Inventory

Build Configurator

Invoicing

Reports

Inventory

Tuesday, July 22

+ New Item

Materials on hand
1,284

Low stock alerts
8

Allocated to jobs
326

Fabric inventory

Live stock across locations

FABRIC	COLOR	LOCATION	AVAILABLE	RESERVED
Linen Blend	Oat	Main studio	42 yd	18 yd
Velvet Luxe	Forest	Warehouse	8 yd	12 yd
Boucle	Ivory	Main studio	26 yd	4 yd
Performance Twill	Slate	Warehouse	51 yd	8 yd

Stock alerts

Actionable material exceptions

Velvet Luxe - Forest
4 yd short for WO-1047

Brass nailheads
Reorder point reached

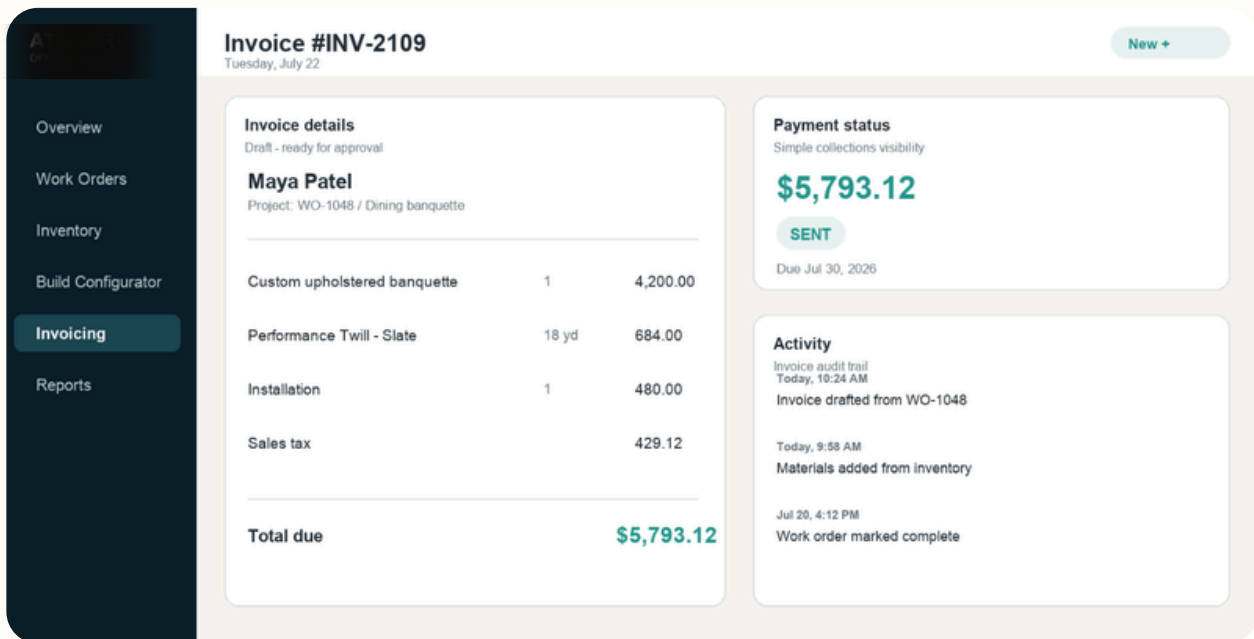
Inventory movement

Last 30 days



Product Screens

Invoicing

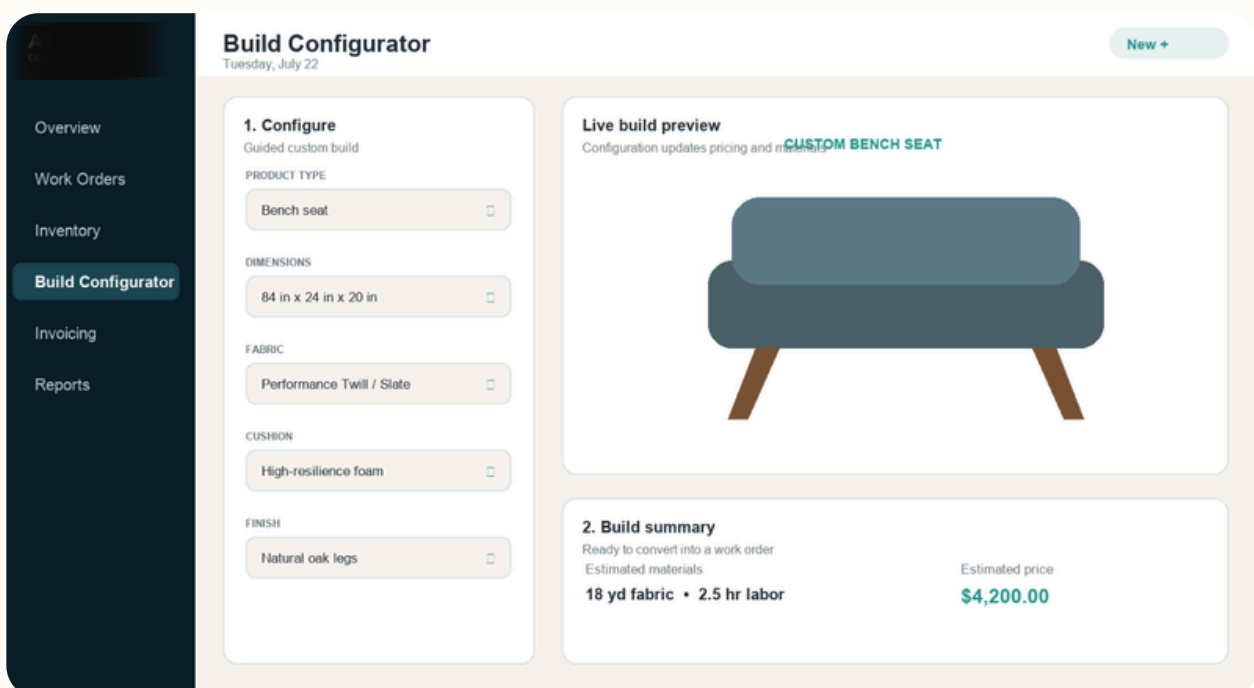


The Invoicing screen displays the details for Invoice #INV-2109, dated Tuesday, July 22. The left sidebar shows navigation options: Overview, Work Orders, Inventory, Build Configurator, Invoicing (selected), and Reports. The main content area is divided into three sections:

- Invoice details:** Draft - ready for approval. Customer: Maya Patel. Project: WO-1048 / Dining banquette.
- Payment status:** Simple collections visibility. Amount: \$5,793.12. Status: SENT. Due date: Jul 30, 2026.
- Activity:** Invoice audit trail. Today, 10:24 AM. Invoice drafted from WO-1048. Today, 9:58 AM. Materials added from inventory. Jul 20, 4:12 PM. Work order marked complete.

Item	Quantity	Price
Custom upholstered banquette	1	4,200.00
Performance Twill - Slate	18 yd	684.00
Installation	1	480.00
Sales tax		429.12
Total due		\$5,793.12

Build Configurator



The Build Configurator screen displays the configuration for a Custom Bench Seat, dated Tuesday, July 22. The left sidebar shows navigation options: Overview, Work Orders, Inventory, Build Configurator (selected), Invoicing, and Reports. The main content area is divided into two sections:

- 1. Configure:** Guided custom build. Configuration updates pricing and materials.
 - PRODUCT TYPE:** Bench seat
 - DIMENSIONS:** 84 in x 24 in x 20 in
 - FABRIC:** Performance Twill / Slate
 - CUSHION:** High-resilience foam
 - FINISH:** Natural oak legs
- 2. Build summary:** Ready to convert into a work order. Estimated materials: 18 yd fabric • 2.5 hr labor. Estimated price: \$4,200.00.

Live build preview: CUSTOM BENCH SEAT





Thank
you!

Collaboration

We appreciate your interest in our case study. At Utkal Labs, we are dedicated to providing innovative solutions that enhance operational efficiency and empower businesses to thrive.



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